

# RICHA LAGHATE

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I am Richa Laghate, a recent graduate, I am motivated and detail-oriented engineer with hands-on experience in product reliability analysis, PCB development, and hardware testing through projects and industrial internships at LG Electronics and BHEL. My aim is to contribute to research and development in the field of electronics and instrumentation by combining theoretical knowledge with practical problem-solving. I thrive in collaborative, innovation-driven environments and am eager to contribute to impactful technological advancements in the electronics domain.

## Education

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| <b>Institute of Engineering &amp; Technology DAVV Indore M.P.</b><br><i>BTech. Electronics and Instrumentation Engineering</i> | <b>2021 – 2025</b><br>CGPA 7.73 |
| <b>Miss Hill’s Higher Secondary School Gwalior M.P.</b><br><i>12<sup>th</sup></i>                                              | <b>2020 – 2021</b><br>81.2%     |
| <b>The Woodstock School Gwalior M.P.</b><br><i>10<sup>th</sup></i>                                                             | <b>2018 – 2019</b><br>85.3%     |

## Professional Experience

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| <b>Graduate Engineer Trainee</b>   LG Electronics, Noida U.P.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>August 2024 – Present</b> |
| <ul style="list-style-type: none"><li>Analyzed field-returned RAC components (ODU/IDU PCBs, remotes, motors, control units) under the Customer Defect Rate (CDR) program to identify root causes based on error codes and service data.</li><li>Performed systematic hardware inspections of PCBs and electronic components to support failure analysis and quality improvement efforts.</li><li>Collaborated in the Development Team on the design and release of New PCBA Development, including contributing to Engineering Change Notices (ECNs) for design and functionality improvements.</li><li>Prepared and analyzed weekly defect reports, identifying trends, non-achieved quality targets, and root causes to support continuous improvement initiatives.</li><li>Participated in New Product Introduction (NPI) activities, contributing to checklist creation, service-related ECOs, and ensuring compliance with manufacturing and service documentation standards.</li><li>Supported cross-functional teams in improving AC reliability through defect trend analysis and recommendations for design optimization</li><li>Maintained technical documentation and reports detailing test results, component issues, and corrective actions.</li></ul> |                              |
| <b>Summer Trainee</b>   Bharat Heavy Electrical Limited ,Jhansi U.P.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>May 2023-July 2023</b>    |
| <ul style="list-style-type: none"><li>Gained practical exposure to the design, manufacturing, and testing processes of power and distribution transformers.</li><li>Studied core components such as windings, core laminations, bushings, and cooling systems used in large-scale transformers.</li><li>Learned about transformer insulation techniques, fault detection methods, and maintenance protocols used in high-voltage equipment.</li><li>Observed assembly-line procedures including coil winding, core assembly, oil filling, and vacuum drying</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                              |

## Projects

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| <b>Air Quality Index Modelling</b>   <i>2nd Year Project</i>                                                                                                                                                                                                                                                                                                                                                                                                                              | <b>April 2022</b> |
| <ul style="list-style-type: none"> <li>Developed a hardware-based system to monitor and model air quality using environmental sensors including MQ135, PMS5003, and DHT11.</li> <li>Programmed microcontroller (Arduino/Raspberry Pi/ESP32) for real-time sensor data acquisition and AQI computation based on national standards.</li> <li>Implemented onboard data visualization using OLED/LCD display to show live air quality metrics (PM2.5, CO2, temperature, humidity)</li> </ul> |                   |

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| <b>Automated Integrated Smart Healthcare Assistance (AISHA)</b>   <i>Smart India Hackathon</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>October 2023</b> |
| <ul style="list-style-type: none"> <li>Designed and developed an integrated smart healthcare assistance system using a combination of hardware and software technologies.</li> <li>Implemented features for automated health monitoring and voice assistance, incorporating sensors and peripherals connected to the Raspberry Pi.</li> <li>Utilized machine learning or AI algorithms running on the Raspberry Pi</li> <li>Enabled real-time data analysis and visualization, providing users with actionable health insights via a user-friendly interface.</li> </ul> |                     |

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| <b>Automatic Door Lock System</b>   <i>C, Assembly</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>April 2023</b> |
| <ul style="list-style-type: none"> <li>Developed a door lock system using a microcontroller for secure access control.</li> <li>Utilized a 8051 Microcontroller, 12V DC motor, Keypad for password, LCD for display.</li> <li>The motor rotates in a clockwise direction to open the gate whenever the password is correct; otherwise, it remains locked.</li> <li>Implemented features for secure locking and unlocking mechanisms.</li> <li>Utilized microcontroller programming languages such as C and possibly Assembly.</li> </ul> |                   |

### Technical Skills

**Skills:** Circuit Analysis, Circuit Designing, Electronics, KiCAD, AutoCAD, Siemens NX,Data analysis, Root cause analysis, MS-Excel, PowerPoint, VLSI, CMOS, Digital Circuits, Analog Circuits, Logic Gates, Testing and Debugging.  
**Languages:** Verilog HDL,C, C++, Python, Assembly, Java, SQL,HTML,CSS,Javascript.  
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### Certificates

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|-------------------------------------------------------------------------|----------------------|
| Certification for CMOS Digital VLSI design   NPTEL IIT Roorkee          | <b>March 2025</b>    |
| Certification for KiCAD PCB Designing   Udemy                           | <b>January 2025</b>  |
| Certificate for Web Development   Saylor Academy                        | <b>January 2024</b>  |
| Certification for Analog & Digital communications   NPTEL IIT Kharagpur | <b>October 2023</b>  |
| Certification for C/C++ programming  Coding Ninjas                      | <b>July 2023</b>     |
| Certificate for Object Oriented programming in JAVA   Coursera          | <b>December 2023</b> |

### Volunteering

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|---------------------------------------------|--------------------|
| National Service Scheme, DAVV B Certificate | <b>March 2025</b>  |
| Academic Cadet Robin Hood Academy           | <b>August 2024</b> |
| Cadet Make A Difference                     | <b>August 2024</b> |